

Debasis Tripathy

Senior Software Developer

AI Engineer & Full-Stack Developer with 3+ years of experience in designing and deploying **agentic AI systems**, **retrieval-augmented generation (RAG) pipelines**, and **scalable backend solutions**. Skilled in building production-ready applications, optimizing infra for millions of users, and combining **academic research** with **real-world engineering**.

✉ debasis.tripathy@rutgers.edu

📍 Harrison, New York, United states

🌐 linkedin.com/in/debasis-tripathy

🗣️ medium.com/@codemass

📞 +1 8622154840

📄 debasistripathy01.github.io/

🐙 github.com/debasistripathy01

SKILLS

Python

Typescript

NodeJs

ExpressJs

MongoDB

Agentic Retrieval-Augmented Generation

PosgresQL

AWS

Operating Systems

Data Structures Algorithms

Low Level Design

ETL pipeline

CI/CD

Large Language Models

Model Context Protocol

WORK EXPERIENCE

Graduate Student Assistant Rutgers University

03/2025 - Present

Newark

Achievements/Tasks

- Taught 2-3 juniors in data structures and data mining.
- Proctored students in New brunswick campus

Software Developer Integra Micro Software Services

11/2023 - Present

Bengaluru, India

Software service, open-source solution service providing company

Achievements/Tasks

- Designed a comprehensive buyer application reconciliation settlement process(RSP) that facilitated smooth payment integration with razorpay, the solution can scale nearly half a million of users and reduced the response time from 6 seconds to 2 seconds.
- Kafka communication protocol setup for the change data capture (CDC).
- **Implemented Redis Caching with High Availability** Configured Redis caching with sharding and indexing across more than two clusters (master-slave architecture) to ensure failover protection and enhance the reliability of an enterprise database.

Contact : Veera Reddy Beereddy (Senior Technical Manager) - veerarb@integramicro.com

Software Developer Antino Labs Pvt Ltd

06/2023 - 09/2023

Bengaluru, India

High growth startup providing B2B services to seed level startups

Achievements/Tasks

- Simultaneously handled two clients (Quicksell and HCL) while managing and mentoring a team of over four junior developers in backend technology.
- Designed and deployed AWS Lambda layers for resizing uploaded documents; developed and scaled APIs to support nearly 2 million users for data sorting and filtering.
- Developed 3 migration scripts for the production database and redesigned the database to enhance users experience in order services. Added 3 basic features to more than 45k users.
- Integrated Socket.io with the i-Dream-whiteboard to enhance real-time communication capabilities.
- Implemented automatic order cancellation and rescheduling features, along with CI/CD setup for UAT and development servers.

Contact : Sani Yadav (VP Technology) - sani@antino.com

WORK EXPERIENCE

Software Developer

Fabzen Technologies Pvt Ltd

03/2023 - 06/2023

Bengaluru, India

Real money game product based company

Achievements/Tasks

- **Led the development of the admin panel** for the Ludo-Empire game, building it from the ground up to ensure seamless management and control.
- Integrated third-party applications for automated user identity verification, enhancing the security and streamlining the user onboarding process.
- Integrated communication APIs including Twilio, Mailgun, and SendGrid for SMS and email authentication, improving the reliability and security of user communications and optimized for 100 request per second concurrently.
- Redesigned the game scheduler to support over 1 million users, migrating from a round-robin system to a scalable queuing system capable of handling 100 requests per second.

Contact : Deepanjal Mitra (Manager) - deepanjal@fabzentechnology.com

Research Assistant

Siksha O' Anushandhan (Research Cell)

03/2021 - 02/2022

Bhubaneswar, Odisha, India

Educational and Research Institute in the field of engineering and medical

Achievements/Tasks

- **Developed Multiple Hybrid LLM Models** : Created advanced models such as IF-EEMD-LSTM, IF-CEEMDAN-LSTM, Boltzmann Machines, RVFLNN, and MK-RVFLNN for predictive analysis and research purposes.
- **Contributed to DST Smart Grid Project** : Played a key role in the Department of Science & Technology's smart grid initiative by developing MATLAB and DSpace models for accurate and efficient simulation.

Contact : Dr. PK Dash (Director Research Consultancy cell) - pkdash@soa.ac.in +91 9437003306

EDUCATION

Information Technology and Analytics

Rutgers University

01/2025 - Present

3.2

Courses

- EEG epilepsy detection with a hybrid machine learning model (globalneurolab.com).
- SIEM security implementation using ELK stack.
- ProResume resume builder for students (proresume.us).

Bachelor Of Technology in Electrical Engineering (Hons.)

Biju Patnaik University of Technology(BPUT)

07/2016 - 08/2020

Berhampur, Odisha, India

Courses

- Electrical Engineering(major - Honors)
- (Major)Temperature, Irradiance and Power Prediction using LLM models for Photovoltaic panel.
- (Minor)Fault detection and protection of transmission lines using LLM models.
- Python, MATLAB, C++, OS certifications.

PERSONAL PROJECTS

ProResume.us (03/2025 - Present)

- Resume website for students

Personal Portfolio Website using React-Redux (10/2022 - 01/2023)

- Developed a personal portfolio with a Redux architecture, integrating GitHub APIs to automatically reflect changes from my GitHub profile on the site. This project showcases my expertise in front-end development and state management, and stands as one of my best works so far.

GlobalNeuroLab.com

- EEG brain data analysis using small LLM models

Security

ACHIEVEMENTS

Research Intern (Indian National Academy of Engineers)
(11/2021 - 01/2022)

Selected among 60 students nationally to research on LLM models with smart grid.

Institution of Engineers Student Member
(03/2019 - Present)

Awarded the student membership for the solar water pump research work submitted by professors.

IEEE Student Member (08/2019 - 08/2020)

PUBLICATIONS

Journals

Power forecasting with Minimal loss using LSTM

02/01/2021

IEEE Journals (International Publication)

Paper represents temperature, irradiance and PV power output prediction with the help of time series predictor Long-Short Term Memory (LSTM)

National Conference Magazine

Long Transmission line Protection by over current relay

Dr Saroj Pradhan

Dr. Lokanatha Tripathy, Debasis Tripathy

20th October 2019

NCGTR-2019

Publication on national conferences magazine

CERTIFICATES

Python programming a practical approach
(07/2020 - Present)

Training by IIT, Kanpur

FDP on Deep Learning a practical approach
(06/2020 - Present)

Training provided by IIT, Roorkee

FDP on Machine Learning and Data analytics
(05/2020 - Present)

Training provided by IIT, Roorkee

LANGUAGES

English
Full Professional Proficiency

Hindi
Full Professional Proficiency

Odia
Native or Bilingual Proficiency

German
Limited Working Proficiency